

Navel - a social robot with verbal and nonverbal communication skills

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ABSTRACT

As robots are used not only in factories but increasingly in everyday life, interaction with them must be as intuitive as possible for everyone. We present Navel, a mobile social robot that supports nonverbal interaction in addition to verbal communication. Thus, the interaction with our novel hardware-software-system is similar to human-to-human communication and appears natural and intuitive for everyone. For social signal perception, multiple computer vision algorithms run in near real-time on an NVIDIA edge device in the robot, achieving low latency and maintaining privacy. For the representation of social signals, an innovative combination of displays and fiber-optic plates is used, allowing the creation of expressive three-dimensional eyes for building realistic eye contact. A lean high-performance software architecture is used to generate agile, lively and context-sensitive behavior. An abstracted humanoid design avoids the Uncanny Valley.

CCS CONCEPTS

• **Computer systems organization**; • **Embedded and cyber-physical systems**; • **Robotics**;

KEYWORDS

human robot interaction, social robot, nonverbal communication, social cues

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1 INTRODUCTION

Robots are increasingly used in everyday life - as service robots, logistics robots, social robots and more.

This implies that also non-tech-savvy and even vulnerable people interact more frequently with robots. Interaction with robots should therefore be as intuitive as possible, i.e. based on familiar interaction patterns. In interpersonal interaction, a complex and situationally adapted interplay of verbal and non-verbal dialogue has been established over millennia. For most people, it would therefore be easiest if robots also mastered these established verbal and

non-verbal forms of dialogue [1], such as making eye contact [2] and continuously signaling their attention and intent nonverbally. Today's robots, however, do not yet have such capabilities.

Research and development of the verbal dialogue capabilities of technical systems is very prominent and has shown tremendous progress in recent years. For a pleasant sympathetic interaction, robots must also be able to perceive and send social signals. There has been great progress in the field of computer vision, making the perception of social signals in real time possible even in mobile systems. On the output side of nonverbal communication, avatars have shown a progress in representing social cues. In the presentation of facial expressions, there are individual projects such as Amica [3], which attempt to reproduce human facial expressions as realistically as possible using mechanical systems. However, such systems are technically very complex, so that they are hardly suitable for mass use. They have not yet managed to cross the Uncanny Valley, since the realistic full imitation of human facial expressions remains a huge challenge. Also, a detailed reproduction of human appearance is not necessary for the respective interaction. So far, there are no robots or other holistic hardware systems that allow interactive intuitive human-like communication with support of nonverbal communication.

The solution presented here integrates state-of-the-art capabilities in real-time social signal detection on an edge device with a novel way of displaying social signals on the developed robotic system. A new hardware solution combines the efficient and simple representation of diverse facial expressions via displays with the possibility to display them three-dimensionally. Thus, for the first time, an interactive software-hardware system can be presented that can communicate non-verbally in real time. Through a deliberately abstracted humanoid design, the Uncanny Valley is circumvented and at the same time the possibilities of sending human non-verbal expression are used for the robot.

This new solution significantly reduces the barrier for non-tech-savvy people to interact with a robot because the integration of non-verbal communication makes the interaction very familiar and intuitive. This enables novel applications in social settings such as elderly care or childcare and allows HRI researchers to explore new interaction strategies.

2 SOLUTION

We introduce Navel, a mobile social robot initially developed within a BMBF research project. Through a deep tech approach, Navel has the capability to perceive and send social signals, which enables a very human-like form of communication. Thanks to the high-performance interaction of perception and representation, Navel can autonomously establish eye contact. The humanoid design is very appealing and sympathetic and is at the same time abstracted enough to avoid the Uncanny Valley [].

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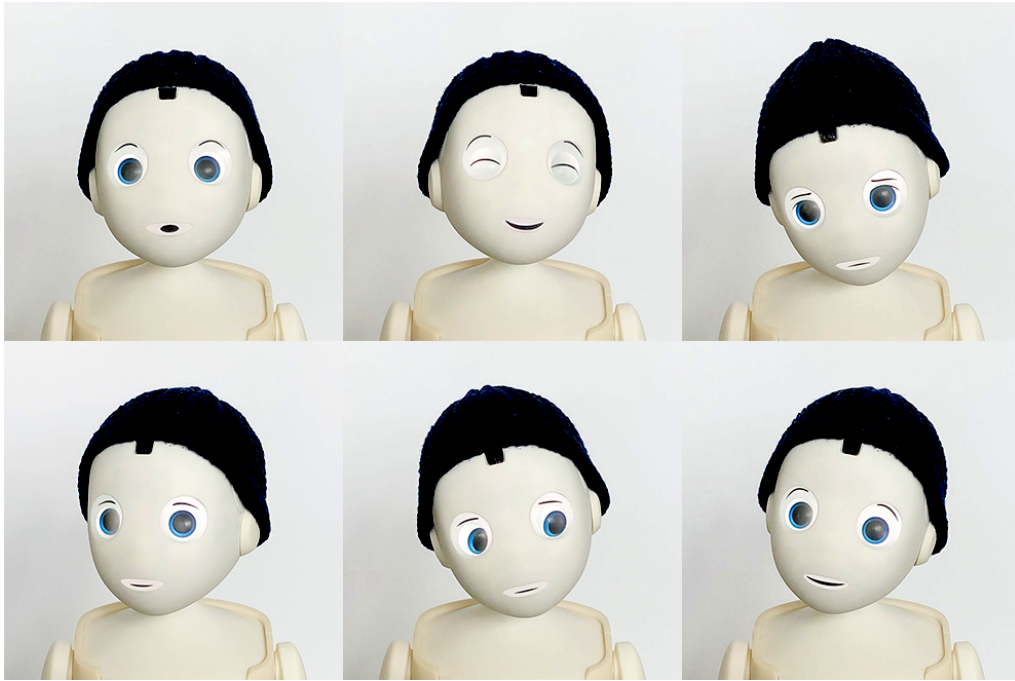


Figure 1: Based on the seamlessly integrated OLED displays and the 3DOF neck, Navel is able to show countless different facial expressions. The six examples presented show expressions that can be interpreted as: 1. amazed, 2. laughing, 3. sad, 4. observing, 5. reflecting, 6. friendly prompting.

2.1 Technical design

Navel has the following sensors for human and context perception: 2 RGB cameras, 3D camera, 2 lidar sensors, 3 sonar sensors, 3D microphone array. As actuators, Navel has a differential drive including tilt mechanism, 3-axis head, 2 shoulder motors to rotate arms, loudspeakers and 3 displays for 2 eyes and the mouth. In front of the displays are fiber optic plates, which project the image of the 2D displays onto the 3D milled surface. Thus, the graphics are seamlessly integrated onto the robot surface and form 3D eye-balls. The fact that the eyes are three-dimensional is a mandatory requirement to be able to distinguish real eye contact from gaze aversion. If eyes are only shown on a 2D display, a so-called Mona Lisa effect [5] is created, as is also known from portrait pictures, for example. In 2D faces you can never exactly estimate the direction of the portrait's gaze and always feel like you are being looked at from every angle.

An NVIDIA Jetson Xavier AGX is used as the main processor, running the backbone to process the sensor data in near real-time. Currently, 5 computer vision algorithms are running at 10 frames per second for face recognition, identity recognition, facial expression recognition (estimating emotion), head orientation estimation and gaze direction estimation. With support of the microphone array, sound sources can be localized and tracked in the three-dimensional space.

Speech-to-text conversion is currently performed online via Google Cloud Speech. For the dialog manager the open source tool Rasa is used and for the speech output an Acapela voice. Lip synchronization is implemented so that the mouth moves with the

speech output. The eye and mouth displays can be used to show the main emotions as well as to interpolate smoothly between them, so that an extremely large number of expressions can be shown. Standard behaviors are implemented that realize a natural lively basic behavior such as continuous small movements of head and facial expressions or gaze aversion during eye contact.

Navel can currently be operated in 3 ways: Via a Python SDK and an API, Navel's behavior can be freely programmed. Via Rasa, dialog-centric interactions can be implemented including non-verbal signals. Navel can also be controlled remotely in real time via a browser interface, e.g. for Wizard-of-Oz tests.

The inner construction is based on a bent-sheet design as well as FDM printed and other mechanical parts. The outer shell of the current Navel prototype consists mostly of SLS 3D printed parts.

2.2 Operation and use cases

Our prototype can currently demonstrate various use cases that can also be shown at CHI23.

Vivid Mode: The robot shows vivid facial expressions and micro-movements. It aligns itself with sound sources. If it detects a face there, the robot establishes eye contact. Instead of just looking rigidly into one eye, the gaze focus alternates between both eyes and the user's mouth, including complete gaze aversion at times, to ensure pleasant eye contact. The Vivid Mode alone, with continuous small movements, reaction to sounds and eye contact, creates the impression that Navel is a robotic being that understands context and has its own intentions.

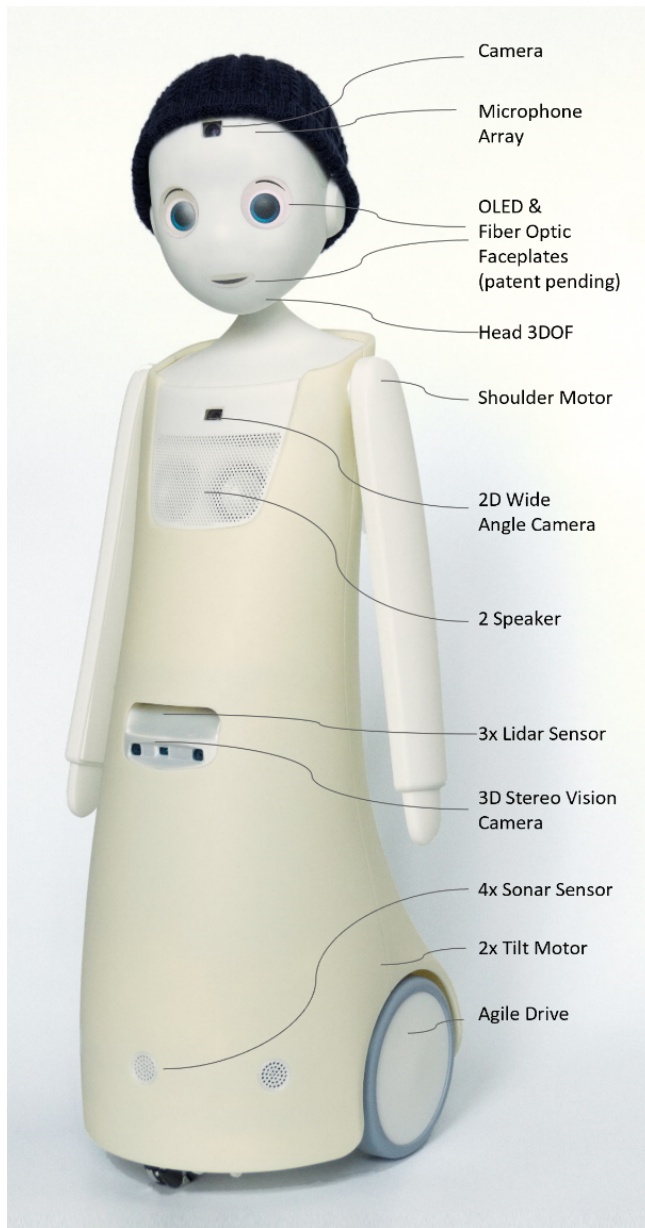


Figure 2: Navel is about 75 cm tall, mobile and equipped with various actuators and sensors. The cap supports the sympathetic design and makes Navel seem less technoid.

Dialog mode: The robot conducts a basic demo dialog with the user. It asks e.g. for the name, the origin and the well-being. It may tell jokes, do a quiz and ask personal activating questions. Non-verbal signals such as facial expressions and voice pitch are also used to reinforce the dialog content and convey a natural impression.

Wizard-of-Oz: The robot can be controlled remotely via a browser interface. The robot's emotions can be set. One can see the camera image of the robot including the evaluation of the neural

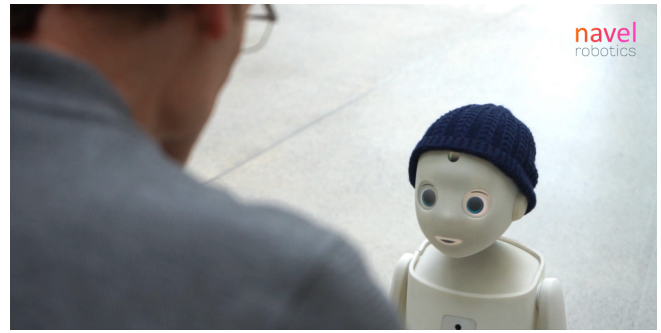


Figure 3: Based on the open source dialog manager Rasa, Navel conducts a dialog with the user. In parallel, non-verbal signals are also perceived and sent by Navel.

networks. The head and gaze orientation can be controlled via this. Small Python scripts can be started, with which a short interaction including verbal utterances with the user (Wizard-of-Oz study) can be carried out. Navel's locomotion can be controlled via a joystick.

2.3 Outlook

The functionalities of Navel are being continuously developed. In the course of the year, for example, the autonomous navigation of the robot in human environment will be optimized. In addition, the hardware is currently being developed towards pilot series, which is to be produced from May.

From mid-2023, the Navel robot will be available for the human-robot interaction research community.

Pilot projects in nursing homes will take place later this year. The aim is to optimize interaction with vulnerable target groups and reliability in field use.

For autonomous interaction in nursing homes, a cognitive layer will be developed to control the behavioral and dialog strategies of the robot, taking into account the context of the users.

3 CONCLUSION & IMPACT

With Navel, we present a new level of interaction with robots. Because Navel can handle non-verbal communication and perceive and send social cues, the interaction is more human-like and therefore more intuitive for users.

These capabilities offer particular advantages in several areas:

Researchers in human-robot interaction can use Navel's new capabilities to explore new forms of interaction, especially in the nonverbal domain.

Navel can also be used with vulnerable target groups due to its strong perceptual capabilities and social skills. For example, the goal is for Navel to autonomously address those in need of care in nursing homes, activating them emotionally and cognitively independently of caregivers. Studies [6] show that, for example, social robots can sustainably improve well-being through positive psychology interventions. If the interaction is to be initiated by the robot, as is the case here, it is crucial that the robot approaches the users very sensitively. To control this process, it is essential that it continuously analyzes the social signals of the counterpart and sends appropriate signals to inspire trust. Initial Wizard-of-Oz

studies have shown that older target groups enjoy interacting with Navel.

Educational/children's applications are also possible. In tests with children with autism, Navel's strong facial expressions appealed to the children very well.

Workers in social professions, which are often understaffed and overworked due to unfavorable key ratios, could get relief and support through Navel.

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